CP193.008 - Project Proposal

Minerva University

Computer Science and Artificial Intelligence

CP193.008 - Capstone Directed Study

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1 Product Summary

1.1 What the Application Does

I am building a platform that explains the Quran. A user asks a question or goes directly to a specific surah (a collection of verses). Each verse appears in Arabic, with a Korean explanation of its context underneath, not a literal translation. The user can toggle English on or off. The platform explains four parts of each verse: the meaning of its words, the historical background, the context, and the other verses it connects to. When a verse depends on other verses, the platform shows them together to give the full picture. Every explanation links to the source it comes from.

The platform explains a word by what it means in its verse, and does not isolate the word and give a dictionary entry. The root ضرب (pronounciation: Darab) shows why. Its literal sense is to hit or beat.[1] In "to set forth a parable" (2:26) it means to present,[2] and in "when you travel in the land" (4:101) it means to travel.[3] A learner who cannot read the Arabic cannot tell which sense applies, so the platform states it for each verse. The user taps a word, sees its meaning in that verse, and expands it when more detail is needed.

Each surah opens with a brief that places it in the Quran and lists the verses it connects to. A search bar returns the closest results and suggestions. A Q&A panel answers questions about a verse and cites the sources behind the answer. The system declines to answer when the sources do not support one. A beginner guide, either an animated walkthrough or a short video, shows first-time users where to start. Learners also get exercises, described in Section 1.2. Figure 1 shows the first sketch of the interface. I replace it with a working interface in sprint one.

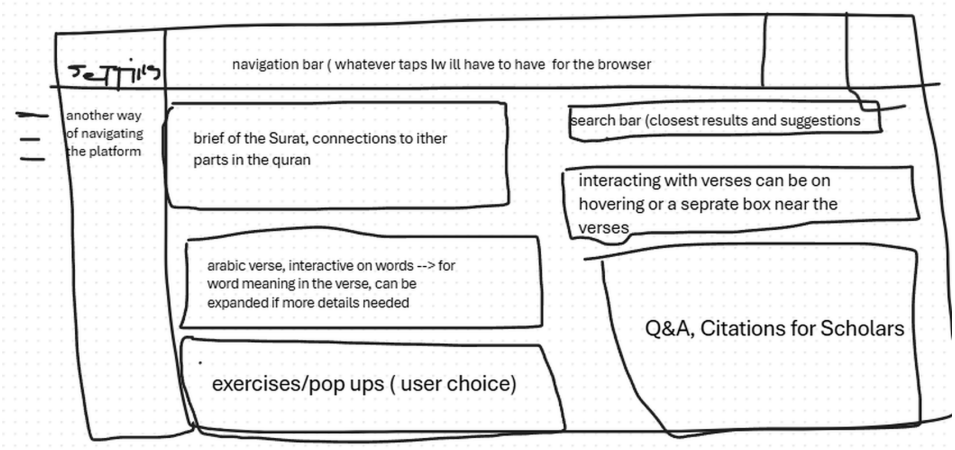


Figure 1: First interface sketch. The verse view holds the surah brief, the search bar, interactive Arabic words, exercises as a user choice, and a Q&A panel with citations for scholars.

1.2 Who It Is For

Two groups use the platform, and each asks for something different.

The first group is anyone who wants to understand a verse, Muslim or not, at any level of knowledge about the religion. Some of them examine one debatable point and want to see where a claim comes from and what supports it. This group needs the Q&A panel and the citations, and skips the exercises.

[1]Quran 2:60 (Al-Baqarah): فَقُلْنَا اضْرِب بِّعَصَاكَ الْحَجَرَ. Moses asks for water for his people, and Allah tells him to strike the rock with his staff. The verb اضْرِب describes a physical strike, which is the root's literal sense.

[2]Quran 2:26 (Al-Baqarah): إِنَّ اللَّهَ لَا يَسْتَحْيِي أَن يَضْرِبَ مَثَلًا مَّا بَعُوضَةً فَمَا فَوْقَهَا. The verse says Allah does not hesitate to give an example, even one drawn from a mosquito or what is above it. The phrase يَضْرِبَ مَثَلًا reads literally as strike an example. Arabic uses it to mean present an example, and no striking occurs.

[3]Quran 4:101 (An-Nisa): وَإِذَا ضَرَبْتُمْ فِي الْأَرْضِ فَلَيْسَ عَلَيْكُمْ جُنَاحٌ أَن تَقْصُرُوا مِنَ الصَّلَاةِ. The verse states there is no blame in shortening the prayer while travelling. The phrase ضَرَبْتُمْ فِي الْأَرْضِ reads literally as you struck in the land. Arabic uses it to mean you travelled.

The second group is the primary audience: Korean-speaking learners who do not speak Arabic, new converts among them. English resources exist already, and the sites I searched offer no Korean explanation of the verses.

Learners differ from each other. A learner who inherited the religion and studies for memorization needs repetition, so exercises serve that learner. A new convert who wants to keep up with what they cover each time needs exercises as well. I will build the exercises in the style of Duolingo lessons, so they are interactive and familiar to learners. Daily and weekly reminders bring learners back to them, and accounts store progress and goals across sessions. A learner with one specific question wants that piece of information to clarify a point or make a decision, and does not want to study everything. A questionnaire at sign-up sets the exercise types for each learner. This way I do not force exercises on someone who came for one answer.

I chose Korean for two reasons. My premise is that a product in the user's native language speeds learning more than the same product in a second language, including for bilingual and multilingual users. The second reason is the lack of resources. During my rotations in Taipei and Seoul, and in conversations with Muslims in South Korea and with my peers here, I learned that the materials available to Muslim communities there do not explain the verses properly. The translations I found transliterate instead of translate, and that changes the context of the verse.

1.3 How It Helps the User

For a Korean-speaking learner, the platform replaces a transliterated translation with an explanation in Korean under the Arabic verse, with the history and context behind it. Reminders and exercises keep the learner on a schedule, and saved progress lets the learner return where they stopped. For a reader who checks a debatable point, the platform gives the answer with its sources, so the reader verifies the claim without searching across books.

The long-term goal is to add more languages. I collect feedback from users at the end of each sprint during the capstone, and I plan to collect it monthly after the capstone ends.

2 Market Research

I looked at five different platforms: Tanzil, the Quranic Arabic Corpus, Almaany, Quran.com, and Athars. I ordered them from least to most useful. This ranking is based entirely on how I went through the five platforms and how easy it was, personally, for me to find the information I needed. I then thought about it from the perspective of my users. There are no stats or ranking at all here. I checked each site for interactivity, language provision, correctness of the content, and ease of use. Two of the five sites have Korean content, Tanzil and Athars, and in both, the Korean is not the correct contextual translation (Quran.com is way much better than Tanzil). Tanzil renders Allah as 하나님 (hananim), the term Korean Christians use for God.[4]

2.1 Tanzil

Tanzil was the least useful of the five. The interface is plain: the Quran text and a side panel for search, browsing, recitation, and translation (Figure 2a). Tanzil presents itself as a provider of accurate Quran text and translations in several languages. It does have several languages, yet the translation is not correct at all.

For interactivity, the user browses by surah, verse, juz' (a chapter in the Quran), or page, plays a recitation, and reads a translation in a fixed box or on mouse over. The user cannot ask a question, and Tanzil gives no explanation of a verse.

For language provision, Tanzil has a Korean translation. The translation is not the correct contextual translation. The Korean text of verse 1:2 renders Allah as 하나님, the term Christians use as well. I back-translated the Korean text with Google Translate (Figure 2c), and it reads "I offer praise to God, the Lord of the entire universe." The English translation on the same site (Figure 2a) reads "[All] praise

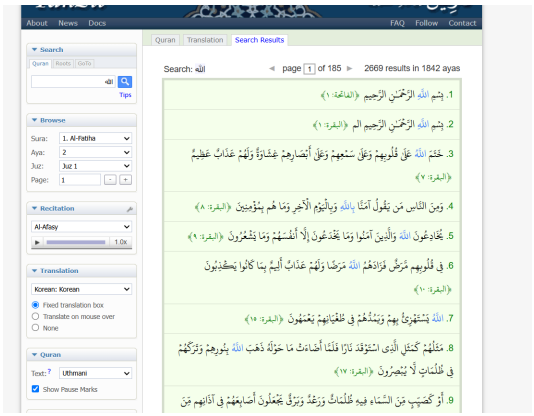
[4] 하나님 (hananim) is the Korean word for God that Christians use and missionaries introduced it in the late nineteenth century.[12] Allah is a name, while God is a general reference that a believer in any religion can use for the being they worship. Using 하나님 "for Allah" therefore gives the verse a Christian connotation.

is [due] to Allah, Lord of the worlds." A Korean reader gets a term Christians use in place of Allah, and the context of the verse changes. The claim of accurate translations does not hold for Korean.

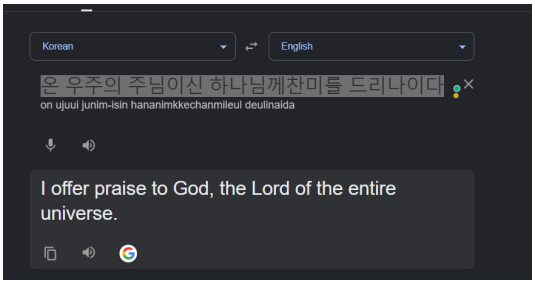
Search works in Arabic only. I searched for a phrase from the Korean translation (Figure 2d) and for Allah. Allah was shown with so many results in Arabic (Figure 2b), but not in English or Korean, and Tanzil returned 0 results in 0 ayas (verses).



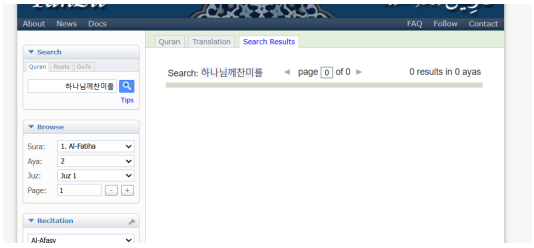
(a) Verse 1:2 with the English translation.



(b) Search for "Allah" in Arabic.



(c) Back-translation of the Korean text of verse 1:2.



(d) Search for "Allah" in Korean.

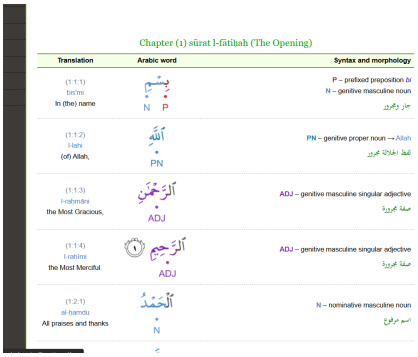
Figure 2: Tanzil Platform exploring.

2.2 Quranic Arabic Corpus

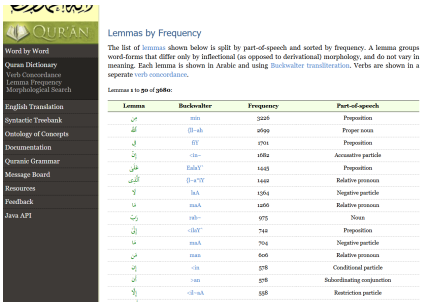
The second source was the Quranic Arabic Corpus. The website is simple and not appealing. It presents itself as an annotated linguistic resource that shows the grammar, syntax, and morphology of each word in the Quran. The menu has word by word, a Quran dictionary, English translation, a syntactic treebank, an ontology of concepts, and documentation. The annotation method behind the content is published and evaluated.

For interactivity, the user picks a chapter and a verse from two menus and clicks the Arabic text of the verse to see the details of each word. The word-by-word page shows the pronunciation of each word, a short English translation, and the grammar of the word in English and in Arabic (Figure 3a). The pronunciation is shown properly. The site explains the grammar of a word and not its meaning. For the word Allah in verse 1:1, the page gives the tag "genitive proper noun" and the translation "(of) Allah," with no explanation of the meaning. The dictionary is heavy on statistics. It lists 3,680 lemmas, each with its frequency and part of speech (Figure 3b). A lemma is the base form that groups the forms of a word.

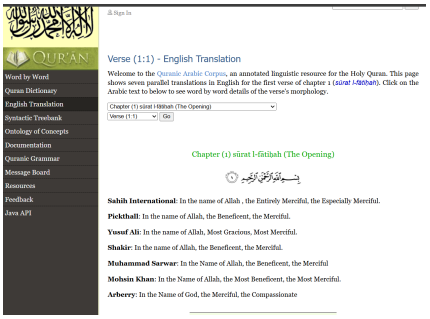
For language provision, the site is in English and has no Korean content. The English translation page shows seven English translations of a verse side by side (Figure 3c). For verse 1:1, Arberry writes God where the other six write Allah, and the two names of mercy are written five different ways across the seven translations. This makes more confusion for whoever wants to learn or just looking for a meaning of any part and still the translation is not implying any helpful infomration. I'd recommend this platform for a learner that has good background or is studying Islamic culture/ Arabic grammar in the context of Qura'n, but not for clarification for someone with no prior knowledge.



(a) Word-by-word breakdown of the verse (pronounciation and grammar).



(b) Quran dictionary: lemmas by frequency.



(c) The different English translations same verse

Figure 3: Quranic Arabic Corpus.

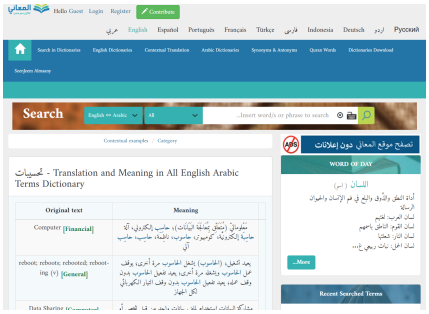
2.3 Almaany

Almaany was the third. The website is a dictionary site, and it looks better than the previous two, UI wise, yet it is still distracting. The pages hold many things that my target audience are not looking for, such as a word of the day, a list of recently searched terms, and a menu of dictionaries (Figure 4a).

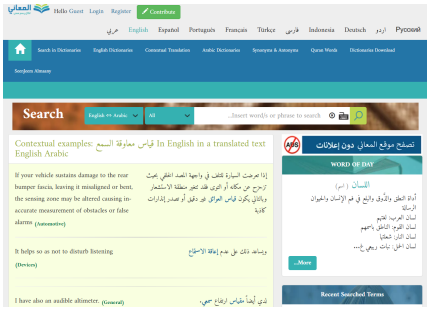
For interactivity, the user searches a word or a phrase, chooses a language pair and a category, and moves through a menu at the header of the site. The menu categorizes the content into different things, and using it also requires knowledge. Contextual translation gives the ability to look up words and phrases, but the content it returns is random and has no explanation at all. It does give a contextual translation, but has nothing to do with a Quran explanation, and it is vague and generic. The examples I got were about the sensing zone of a vehicle bumper and about hearing devices (Figure 4b).

Almaany has eleven languages, and Korean is not among them. Also, when setting the language to English, the Quran words page of the word العالمين is shown mostly in Arabic (Figure 4c). The analysis of the word, its meanings in the Quran, and the tafsir (explanation) of the verse are in Arabic. English appears only in a short meaning of the word, "of the universe," and in a list of English meanings of its root.

The page still shows a literal translation. The user interface is better than the previous two, but it is still not helping in getting a clear meaning.



(a) Dictionary page and menu with the language set to English.



(b) Contextual translation examples unrelated to Qura'n explanation.



(c) Arabic explanation for verses even when selecting different langs.

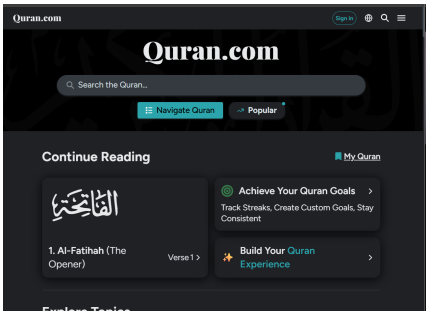
Figure 4: Almaany platform.

2.4 Quran.com

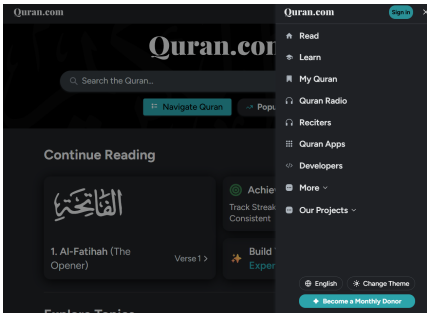
Quran.com is the fourth platform. It beats the previous three by far. The site starts clearly with a search bar for semantic search. Under the search bar, the user has a button to navigate the Quran, a popular section, and a Continue Reading card. It has personalized goals for each user, where they can track streaks and create custom goals (Figure 5a). The site has an index for the surahs, an audio feature, and a learning tab in the menu (Figure 5b). The learning tab holds learning plans, not as exercises or interactivity between the user and the content, but more as pre-made lessons about different surahs in the Quran, each with a specific purpose or story (the Elephant surah, for example, speaks about the king who tried to demolish the Kaaba, the House of Allah; he was stopped easily and was

punished immediately in this life by being thrown with stones that are from hell, and also the elephants he brought to destroy the building refused to move at all) (Figure 5c). The learning plans have some surahs in English and then in Persian, but they do not cover all topics.

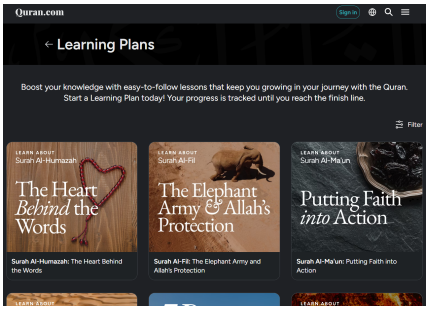
For language provision, Quran.com supports many languages, and the language setting sits at the bottom of the menu (Figure 5b). Korean is not among them. The languages there are only for verses to be read and translated, but when it comes to explanations, it goes back to Arabic and English, so we are facing the same issue again when it comes to explanations.



(a) Home page with the search bar and the goals.



(b) Menu with the learning tab and the language setting.



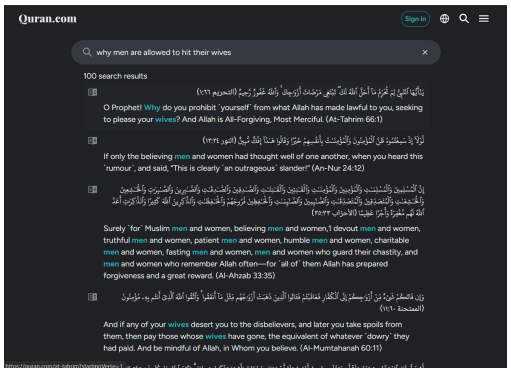
(c) Learning is not interactive but more narrative and static.

Figure 5: Quran.com: what works.

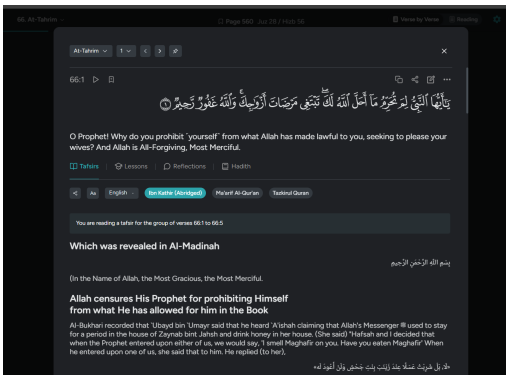
The flaws start with the search. The search is good, but the results it retrieves deviate a lot from a direct answer to the question. I searched "why men are allowed to hit their wives" and Quran.com returned 100 results. The first four results are 66:1, 24:12, 33:35, and 60:11. Each result highlights words from my question, such as why, men, and wives, and none of the four answers the question (Figure 6a). I chose this question for all of the platforms, given that I knew where non-Muslims get the verse from and also the other resources they use to prove their point, and I did enough research on this, so I needed to see how accurately these platforms can tackle this question.

The content is still a direct translation. Quran.com does provide tafsir, which is an explanation of the verse, and for verse 66:1, the user picks between three books: Ibn Kathir (Abridged), Ma'arif Al-Qur'an, and Tazkirul Quran. Nothing in the tafsir is cited. The page lists the scholars and authors, but this doesn't help my users, who have tons of books to look at. The tafsir shown is for the group of verses 66:1 to 66:5, and not for the single verse (Figure 6b).

Quran.com beats the previous three, yet it gives neither a Korean explanation of a verse nor a source the user can check (whether it's Korean or the other languages, it's only Arabic/English for explanation).



(a) Search results for the question.



(b) Testing the accuracy of explanation.

Figure 6: Quran.com: the flaws.

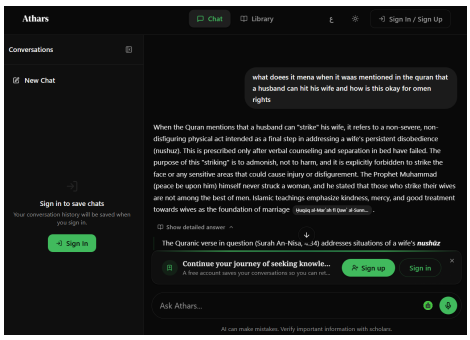
2.5 Athars

Athars is the fifth platform and it beats all the previous apps. It is a chatbot that uses a model which takes a question in any language through semantic search and returns a detailed answer with cited sources in the same language. I learned that it takes any language after my meeting with its developer and not before. Athars was made by an upperclassman, Ali, an M26. I met with him and explained my goal,

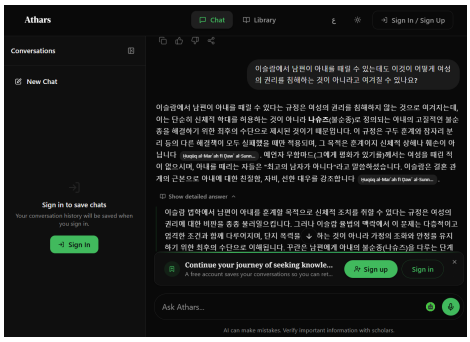
and he agreed to provide an API for the model and explain the model in more detail for me. At the next meeting, I need to have a layout of the platform so that he can see whether his model fits here or not. If it fits, I can modify it because, despite it being the best platform compared to the previous ones, it requires a few more steps, as I mention below, to make sure it's more precise.

For interactivity, the user types or speaks a question, and the answer appears in the chat. Each answer has a short version and a detailed version that opens under "Show detailed answer" (Figure 7a). The detailed answer is broken into sections, for example the conditions and limitations on striking (Figure 10a). The user can also browse the library, and the chats are saved only when the user signs in. Unlike Quran.com, Athars cites its sources inside the answer. The citation is a chip with the title of a book and the page number, and when using the link, it will take the user to the direct part used in the explanation. Yet the books are in Arabic, and this is one of the downfalls of the bot.

For language provision, I asked the same question in English and in Korean (Figures 7a and 7b). The answer comes back in the language of the question. The results were good in English. The results were good in Korean, but still weaker than in English.



(a) Answer to the question in English.



(b) Answer to the same question in Korean.

Figure 7: Athars: the same question in English and Korean.

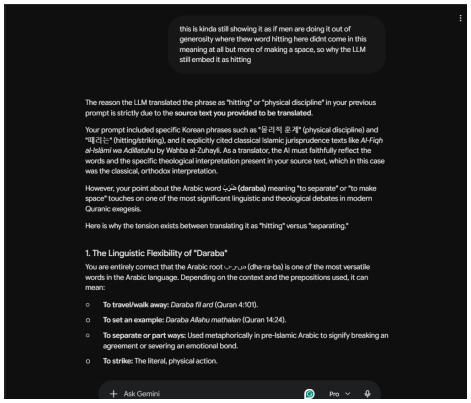
For correctness, both answers explain the verse, An-Nisa 4:34, through the meaning strike. The English answer uses the word "strike," and the Korean answer uses 때릴, which means hit. Both are followed by an explanation from books of fiqh.[5] The English answer lists the conditions of the act: non-severe, no strikes to the face or sensitive areas, limited tools, and a last resort (Figure 10a). In the parts I captured, neither answer shows the other meanings of the root ضرب, the example I used in Section 1. The user gets one meaning of the word and an explanation built on it, and not the range of meanings with the source of each.

I searched for this issue and asked Gemini. It was a short conversation, and I linked it here already: **Gemini chat**. This made it clear for me, but while implementing, I need to be aware of these parts, run it multiple times to see if I made the fixes or not, and get more understanding of what Gemini was suggesting.

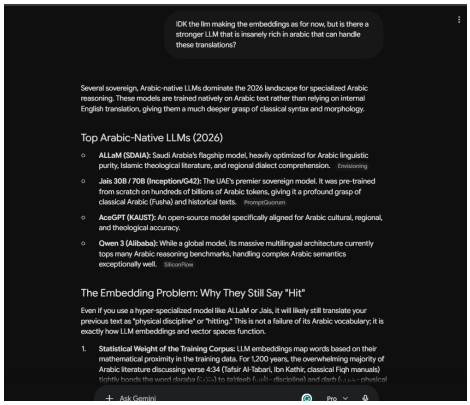
The model returns one meaning because of the weight of the training corpus, not because it lacks the Arabic. Most of the Arabic literature on verse 4:34 ties ضرب to discipline and to the physical act, so the embedding of the question lands there, and a model trained natively on Arabic lands in the same place (Figure 8b).

Then the model returns the root of the word, which is fine, but not the implicit meaning in the context. Gemini returns the other meanings with verses: to travel, in 4:101, to set an example, in 14:24, and to separate or part ways (Figure 8a). These are the meanings Athars did not show.

[5]Fiqh (الفقه) means understanding. In language, it is the understanding of a thing and knowledge of it, and understanding the exact meaning a speaker intends. In the technical sense, it is العلم بالأحكام الشرعية العملية المكتسب من أدلتها التفصيلية, which is the knowledge of the practical rulings of the Sharia, acquired from their detailed evidences. "Of the Sharia" means taken from the Quran and the Sunnah. "Practical" means related to the actions of accountable people, such as prayer and transactions, which excludes the rulings of creed. "From their detailed evidences" means derived from the evidence of each issue on its own.



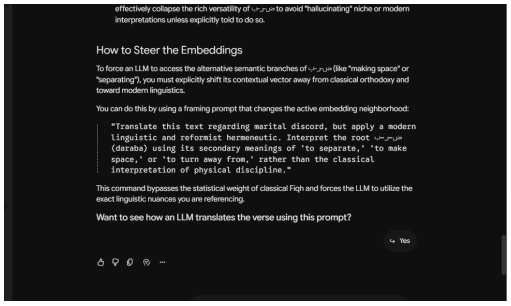
(a) The meanings of the root with their verses.



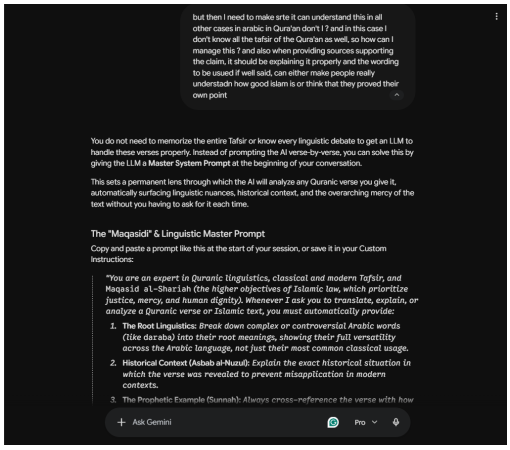
(b) Arabic-native models and the embedding problem.

Figure 8: Gemini: why the one meaning comes back.

Gemini offered some options to fix the output by steering it with a framing prompt that names the secondary meanings of the root and asks for a modern linguistic reading instead of the classical one (Figure 9a). A system prompt set once at the start of the session does the same for every verse, and it fixes what each answer must contain: the root and its range, the reason for revelation, the prophetic example, and named references (Figure 9b). This will highly improve the retrieved answer, but I still need to check if it's applicable, and I need to search on this part more.



(a) Framing prompt that steers the embeddings.



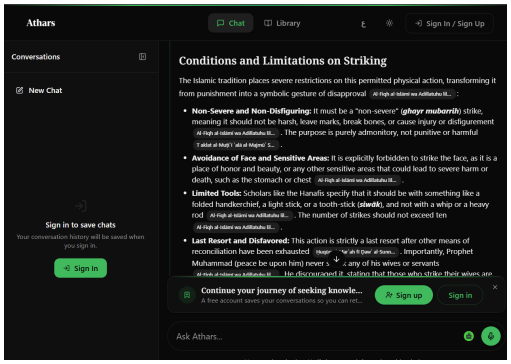
(b) System prompt that fixes what each answer contains.

Figure 9: Gemini: how to steer it.

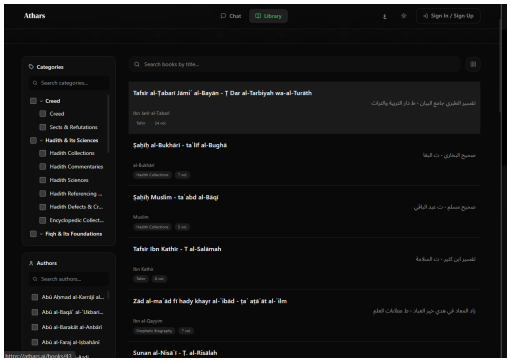
The last thing I asked for was the alternatives. If the Athars API does not reach the quality I need, the chat names Arabic-native models I can test: ALLaM, Jais, AceGPT, and Qwen 3 (Figure 8b). I need to double-check these resources. Also, the extra part I can get from this model is to translate the content of the provided sources into the users' language as well, the bonus point over what Athars is making at the moment.

Gemini is still not a reliable source to confirm the strength of the translation. I use it as a lead on the retrieval behavior, and not as evidence about the meaning. I sent the Korean answer to a Korean Muslim to confirm the wording of 때릴, and I have not heard back yet.

The other good part is the vast library of resources I can use from the bot. The library lists the books by category, author, and title, and it holds the tafsir of al-Tabari and of Ibn Kathir, Sahih al-Bukhari, and Sahih Muslim (Figure 10b), which is what we discussed in the meeting: that I can save more time on getting the needed content (and it would have helped more if they were in English, but I couldn't find proper ones).



(a) Conditions and limitations in the detailed answer.



(b) Library of classical texts.

Figure 10: Athars: the detailed answer and the library.

Athars is highly recommended for anyone looking for information, but it still lacks the learning part, which is what I'm working on. I haven't checked all languages, but I'd say it's doing a decent job on the ones I tested. Yes, its Korean is weaker than its English, the books it cites are in Arabic, and its answers give one meaning of the word, but it can still be used to an extent, or I can make it better.

Table 1 summarizes it all: user interaction, language provision, correctness of the content, and ease of use. Looking up each app, I can pinpoint one thing that I'm planning to implement or build upon their products but with improvements as well.

Table 1: The five platforms compared, from least to most useful.

Platform	Interaction	Language provision	Correctness	Ease of use	What my application does differently
Tanzil	Browse by surah, verse, juz' or page. Recitation. Translation in a box. No questions and no explanation.	Has Korean, but renders it in different meaning (하나님 means God that Christian use). Search works in Arabic only.	The Korean is not the contextual translation, and the context of the verse changes.	Plain interface. Easy to navigate, not useful tho	A Korean explanation of the verse and its context, not a literal translation.
Quranic Arabic Corpus	Pick a chapter and verse, click the Arabic for word details. Dictionary of lemmas and their frequencies.	English only, no Korean. Seven English translations side by side adding more confusion.	Explains the grammar of a word, not its meaning. The translations disagree, for example God and Allah.	Simple, not appealing.	States the meaning of a word in its verse.
Almaany	Search a word or phrase, choose a language pair and a category.	Eleven languages, no Korean. The Quran words page is mostly in Arabic with the language set to English.	The contextual translation is unrelated to Quran explanation.	Better interface, still distracting.	A Korean explanation tied to the verse.
Quran.com	Semantic search, goals and streaks, audio, learning plans as pre-made lessons.	Many languages for verses, no Korean. Explanations only in Arabic and English.	Search results do not answer the question. The tafsir is not cited and covers verses 66:1 to 66:5, not one verse.	Clear start page. Semantic search upon different topics, need more precision but works on general claims	A Korean explanation with a link to its source. Exercises as a user choice.

Platform	Interaction	Language provision	Correctness	Ease of use	What my application does differently
Athars	Chat by typing or speaking, short and detailed answers, library, chats saved on sign-in.	Any language. Korean is weaker than English.	One meaning of the word, not the range. Cites books with page numbers, and the books are in Arabic.	Straighforward Chatbot, familiar UI for anyone used any LLM before, give a promopt and it applys RAG on the books it has on the libraray.	States the meaning of a word in its verse with the source. Adds exercises, reminders, and saved progress. Provide the explaantions in the users' language as well.

3 Implementation Plan

3.1 Scope

The first sprint builds the base of the platform and a first version for users to react to. It will be a login and signup page in English and Korean, with authentication that checks whether the user is logged in. I set up the database for users and use it for login and signup only. Then I build the first few pages of the application, where the user can see a verse in Arabic, Korean, and English with explanations that are verified by language experts first. The feedback on them decides how I improve them in the sprints that follow. Also, as mentioned, the sources are in English for now, and a few of them are enough for this sprint. Each source link opens a page inside the platform that shows the source. The Arabic ones I will review with the SF community here, and then I will reach out to Koreans here for validation as well. I will make it on a small sample, just to give users a glimpse of what they are expecting to find in the platform. The verse page holds fixed questions with prepared answers, so testers can follow the format of an explanation and its source before the model exists. The prompt bar (semantic search) appears on the page and does not work for now, until I get to know more about what model I will use and implement.

I will hold off a bit on semantic search until I finalize it, and this part needs a lot of work. I have notes in my notebook but need to elaborate on them more. I learned about RAG, and I have not worked with a model as large as the one behind Athars, for example, or the models Gemini suggested. I get the model from Ali (M26 Alumni) or find another one, and I run trials in week 7 before the search bar works. The exercises wait as well. I have not decided whether they follow a fixed learning plan or follow the topic that the user researches, and the feedback from the first two rounds decides it. Building them now means rebuilding them if the feedback points to the other option.

3.2 Plan

The sprint runs from week 5 to week 9. A basic process here is the feedback from both the users and my advisors to make sure I'm on the right path, and to see what I missed. I also need to be involved in the testing. We discussed that feedback is one way, and the other way is me observing the interaction of users with the product myself. Table 2 lists the timeline.

Week	Work	Hours
5	Requirement elicitation, content validation, and a short initial questionnaire on what people look for. The questionnaire is not feedback on the product.	8–10
	UI basics and database setup.	6–8
6	Set up the mock lessons and the fixed lessons, check the translation, and record a walk-through video. Share the interface and the current state of the content, and collect feedback. Time is a bit variable here, depending on the users looking at the initial outcome, and then I work from it along with what I planned already.	**12–15, rough estimate**
7	Optimize the model for better context explanation, and run semantic search trials. Collect the second round of feedback through organizations, clubs, and my personal network.	**10–15**
8	Set the exercises, either as static exercises attached to a learning plan or as follow-ups for a quick question, and check that the user is learning.	**8–10**
9	Finalize the platform and run a stability test.	**10**

Table 2: The five weeks of the first sprint.

This table is still an estimate based on how I see the process. I'm still thinking there will be overlapping steps. For example, I was thinking that having the exercises in week 8 is a bit late, so I might make them smaller in scope, so that instead of being surprised in week 8, I can figure things out earlier.

Two courses have work due inside the sprint, CS156 on October 15 and IL18105 on November 30. It should be fine, but there is still space for unexpected events, so my pace might differ. In this case, I will make sure to inform both my advisors ahead of time. Steps that slip because of other assignments move to week 9, which is meant for polishing. The questionnaire in week 5 takes more time than the hours above, because it depends on how fast people reply.

3.3 User Testing

Testers receive the interface I build in week 5 with the content in its current state: the login, the verse page with its fixed questions and prepared answers, the mock lessons, and a demo video that walks through the platform. Round one is in week 6. Round two is in week 7 through organizations, clubs, and my personal network, and my advisor gives feedback in the same week. I will either be with users when running the tests(mostly in the mosque) or requesting, if they don't mind, to share screen or record at any time and then send it to me via email. I need to keep them informed of which elements are static, such as the search bar.

I ask for general feedback first: what is good and what is missing. Then I ask about two sides, the user side and the educational side. Table 3 lists the topics under each. I am not worried about comprehension, because I plan guides in every part of the platform and a starter walkthrough video.

General	User side	Educational side
What is good	Usability	Correctness of the content
What is missing	Persistence	Meeting the requirements
	Adaptability	Enough teaching
	Availability	
	Interactions	

Table 3: What the forms should be covering in a mix of range questions and open ones

The bad part in the feedback/validation from Koreans is the slow pace of replies. I reached out to koreaislam.org and kicaislam.org, and I have not heard back yet. I did get replies from Koreans in SF, so I will keep emailing them and try to see other alternatives, such as a translation center, if there is one.

4 Learning Outcomes and HCs

I made this map after breaking the product into different areas of focus: some technical ones, the data and market one, and global ones. By global, I mean the ones that apply across the whole capstone, and I could not find a better name for them. Figure 11 shows the map, and Table 4 explains how I apply each one.

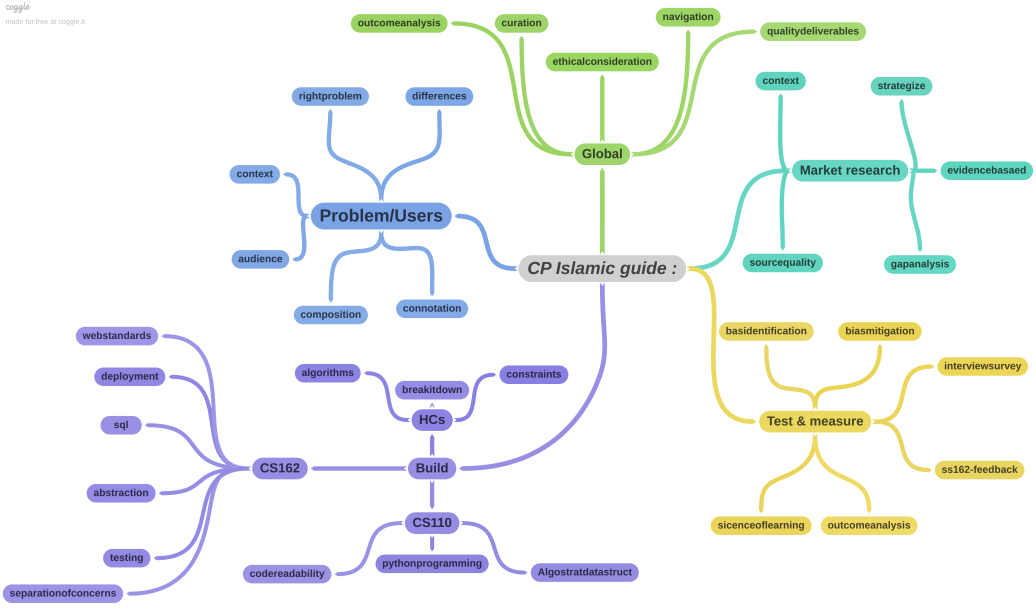


Figure 11: Map of the learning outcomes and HCs I apply, by area of the project.

Table 4: Capstone LOs, course LOs, and HCs, and how I apply each.

Name	How I apply it
Capstone LOs	
#outcomeanalysis	I use the same measures in every round of testing, listed in Table 3: what is good and what is missing, the user side, and the educational side. The same measures in week 6 and week 7 show me whether a change helped. Correctness of the content comes first, because the platform explains a religious text, and a smooth page with a wrong explanation fails its purpose.
#curation	The platform shows sources, so choosing them is part of the product. For this sprint I choose a small sample that I can validate with language experts and the SF community, and each source opens on its own page inside the platform. Every explanation links to the source it comes from, unlike Quran.com, where the tafsir covers verses 66:1 to 66:5 as one block.
#navigation	The plan in Table 2 leaves week 9 for the steps that slip because of CS156 on October 15 and IL18105 on November 30, and I tell both advisors early if my pace changes. For the slow replies, I keep emailing the Koreans in SF, and I look for a translation center or talk to my professor. If Ali's model does not fit, I test the Arabic-native models from the Gemini chat.
#qualitydeliverables	The deliverable of this sprint is a first version that a first-time user can follow without me next to them, which is why I plan guides in every part of the platform and a walkthrough video. The content is verified before testers see it, and the stability test in week 9 checks the platform before I call the sprint done.
Course LOs	
#webstandards (CS162)	One verse page shows three scripts in two directions: the Arabic runs right to left, and the Korean and English run left to right. The login page also switches between English and Korean. Following web standards keeps these pages readable on the browsers and devices that testers use, since I do not control their setup.

Name	How I apply it
#deployment (CS162)	Round two reaches testers through organizations, clubs, and my personal network, so they need to open the platform from a link without me. I deploy the first version before round one in week 6, and the stability test in week 9 runs on the deployed version and not on my machine.
#sql (CS162)	I set up the database for users in week 5 for login and signup. The accounts in Section 1 also store progress and goals, and the questionnaire at sign-up sets the exercise types for each learner, so the users table needs room for those tables in later sprints.
#abstraction (CS162)	The model is not decided yet. It is Ali's or another one. The prompt bar talks to the model through one interface, which in sprint 1 returns the prepared answers. When I choose the model in week 7, the pages stay the same and only what sits behind the interface changes.
#separationofconcerns (CS162)	The interface, the login, the content, and the model each stay in their own part. This matters in sprint 1 because the verses, explanations, and sources are still being validated by reviewers and change often. I can correct them without touching the pages or the login.
#testing (CS162)	Testing happens at three levels in this sprint. The first is the content: the language experts verify each verse, explanation, and source before testers see it. The second is the users: round one in week 6 and round two in week 7 test the platform with the measures in Table 3. The third is the platform itself: the stability test in week 9 runs on the deployed version, and in week 7 I run the 4:34 question through the model options and check which answers hold. Login and signup get their own check, since the pages depend on knowing whether the user is logged in.
#pythonprogramming (CS110)	The model side of the platform is code and data work more than page work. In week 7, I prepare the validated sources, send questions to the model, and compare what comes back across the trials, starting with the 4:34 question I used on all five platforms. I use Python for this.
#algostratdatastruct (CS110)	Semantic search ranks passages by how close they are to a question, so the way the passages are stored and compared decides whether the right source comes first. In the week 7 trials, I compare options on the same questions, and the point where the system declines to answer comes from those results.
#codereadability (CS110)	Ali needs a layout of the platform to decide whether his model fits, and my advisors give feedback on my work, so other people have to read it. I write the code and name its parts so that someone who did not build it can follow how the pages, the database, and the model connect.
#ss162-feedback (SS162)	Feedback decides what the next sprint builds. Round one in week 6 and round two in week 7 use the same measures from Table 3, and I watch testers use the platform as well as read their answers, because what they say and what they do can differ. The feedback from the two rounds decides whether the exercises follow a fixed learning plan or the topic that the user researches, and it decides how I improve the first pages. The feedback from my advisors in week 7 and the validation from the Koreans in SF add to it.
HCs	
#rightproblem	Five Quran platforms already exist, so the problem is not the lack of a platform. The problem I found is that a Korean-speaking learner gets no explanation in Korean with the context of the verse and a source to check. The questionnaire in week 5 checks this before I build the search, by asking what people look for.
#differences	Learners differ from each other. One inherited the religion and studies for memorization, a new convert wants to keep up with what they cover, and a third has one question and does not want to study. The questionnaire at sign-up sets the exercise types for each, so the one with a question does not get exercises.
#context	A word means what its verse makes it mean. The root ضرب means strike in 2:60, present in 2:26, and travel in 4:101, so each verse page states the sense that applies and gives the historical background. Without this, a reader of 4:34 gets one meaning, which is what happened on Athars.

Name	How I apply it
#audience	The main audience is Korean-speaking learners with no Arabic, new converts among them. The explanation is in Korean and not a translation of Arabic or English terms, and the beginner guide and the walkthrough video start from zero. A reader who checks a debatable point is the second audience, and the source links serve that reader.
#composition	The verse page puts the Arabic verse first, the Korean explanation under it, the English toggle, and the source link, as in Figure 1. Round one tells me whether testers find the source link and the English toggle without help.
#connotation	A word carries more than its dictionary meaning. Tanzil renders Allah as 하나님, the word Korean Christians use for God, and this shifts the verse toward another religion. I check the Korean terms with the Korean reviewers before they go on a page, and 때릴 in the Athars answer is the first one I check.
#sourcequality	I vet the sources the platform shows and the sources this proposal uses. On the platform, each source page shows who wrote it, unlike Quran.com, which lists scholars without citing inside the text. In the proposal, I tested the platforms myself and treat Gemini as a lead on how retrieval behaves, not as evidence about a verse. The fiqh definition comes from Dorar and Islamweb.
#gapanalysis	Table 1 compares the five platforms and shows what each one leaves open. Sprint 1 goes after the gaps the verse pages can show: the Korean explanation and the source on every explanation. The exercises, which none of the five has, wait for the feedback.
#strategize	Athars is the closest platform, and Ali offered its model, so I build on it where it is strong, the retrieval and the library, and add what it lacks: Korean, the range of meanings of a word, and the learning part. My weaknesses are five weeks and no experience with a model of that size. The plan answers them with Ali's help, validation from the Korean community in SF, and the Arabic-native models as the fallback.
#evidencebased	I chose one question, why men are allowed to hit their wives (4:34), for all of the platforms, so the comparison rests on the same case and the screenshots show the results. The platform follows the same rule: every explanation links to the source it comes from, and the system declines when the sources do not support an answer.
#biasidentification	Athars gives one meaning of ضرب because most Arabic literature on 4:34 ties the word to the physical act. The Gemini chat also showed that my own prompt pulled the translation the same way, since the Korean phrases in it already said hitting. I look for bias in two places: the answers of the model and the wording of my own questions.
#biasmitigation	In the week 7 trials, I run the 4:34 question with and without the framing prompt and the system prompt from the Gemini chat, several times each, and check whether the other meanings of the root appear. I judge the results against the sources, so the fix does not swap one bias for another.
#interviewsurvey	I use two instruments. The short questionnaire in week 5 asks what people look for, before I build. After the sessions, the feedback asks about the platform and what the tester learned. I also watch testers use the platform, because what they say and what they do can differ.
#scienceoflearning	The exercises follow the style of Duolingo lessons, with daily and weekly reminders to bring the learner back. A learner who studies for memorization needs repetition, and a convert reviewing what they cover needs it too. In week 8, I set the exercises and check that the user is learning.
#breakitdown	The map splits the product into areas, and the sprint splits into five weeks. Semantic search splits further: choose the model, store the passages, retrieve, check that the sources support an answer, then answer or decline. Sprint 1 holds only the last step, with prepared answers and their source, so testers react to the format before the model exists.

Name	How I apply it
`#algorithms`	The retrieval and the refusal rule are an algorithm: take the question, find the closest passages, check whether they support an answer, then answer with the source or decline. I write the steps out before I code them, and I test the decline rule with a question the sources do not cover.
`#constraints`	Five weeks, CS156 on October 15, IL18105 on November 30, a model I have not worked with at this size, and slow replies from reviewers and organizations limit this sprint. These are the reasons semantic search and the exercises stay out of the version testers see, and the content stays a small sample.
`#ethicalconsiderations`	The platform explains a religious text to people who trust what it says, so a wrong explanation costs more than a missing feature. The 하나님 on Tanzil shows how one word choice changes a verse. For this reason the language experts verify the content first, every explanation links to its source, and the system declines to answer when the sources do not support one. The database holds only what login and signup need.

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AI Statement

I used Claude (Anthropic) while writing this proposal. I used it to write the LaTeX for the tables and figures. I used Gemini as I mentioned above for understanding the deviation in the translation despite the the higher accuracy in English and then linked it up as well. I treat it as a lead on how retrieval behaves, not as evidence about the meaning of a verse. I still need to verify the models it suggested. I used Google Translate to back-translate the Korean text of verse 1:2 on Tanzil.